WEEK 2
PROJECT REPORT

Network Reconnaissance & Zenmap-Based Network Scanning

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DOMAIN	Cybersecurity
PROJECT WEEK	Week 2

Report structure

ELECTIVES - Week 2 Project Module: Footprinting with Multiple Kali Tools (six tools on networkwalks.com)

ESSENTIALS - Week 2, Project Module: Zenmap-Based Network Scanning

Prepared from the two practical-work documents supplied for this report. Screenshots are retained as supporting evidence, and findings are presented as recorded in the source materials.

1. Executive Summary

This combined report presents two Week 2 cybersecurity practical activities. The Electives component documents footprinting and reconnaissance against networkwalks.com using six Kali tools: **WHOIS, WhatWeb, nslookup, curl, wafw00f, and dnsrecon**. The Essentials component documents local network identification and Zenmap host discovery on the 192.168.0.0/24 LAN.

The Electives exercise records domain-registration details, website technology indicators, DNS resolution and enumeration results, HTTP response headers, and a detected web application firewall. The Essentials exercise reports three live hosts on the local subnet.

2. Scope and Approach

The report is based on the supplied Find your local IP address document and the supplied Network Reconnaissance Report. The report preserves the observations and screenshots from those sources. It does not represent a new scan, independent verification, exploitation attempt, or full security audit.

The Essentials activity concerns a private local network. The Electives activity concerns publicly observable information for networkwalks.com. The two activities are kept separate to distinguish local host discovery from external domain and web footprinting.

PART A - ELECTIVES

Week 2 Project Module: Footprinting with Multiple Kali Tools (six tools on networkwalks.com)

The objective was to practise footprinting and reconnaissance using multiple Kali Linux tools to collect publicly observable information about the target website **networkwalks.com**.

Target: https//: networkwalks.com
Observed public IP: 192.232.216.135
DNS resolver used in the exercise: Google Public DNS (8.8.8.8)

13. WHOIS Enumeration

WHOIS was used to review domain-registration information, including registrar, registration dates, registrant details where publicly available, name servers, domain status, and contact information.

- Registrar recorded as GoDaddy.
- The notes state that the domain was first registered in November 2019; the expiry/payment date was not fully recorded.
- Owner details were hidden behind privacy protection.
- The output displayed name servers and domain-status locks.
- The notes explain clientTransferProhibited, clientUpdateProhibited,

clientRenewProhibited, and clientDeleteProhibited as restrictions on transfer, update, renewal, and deletion respectively.

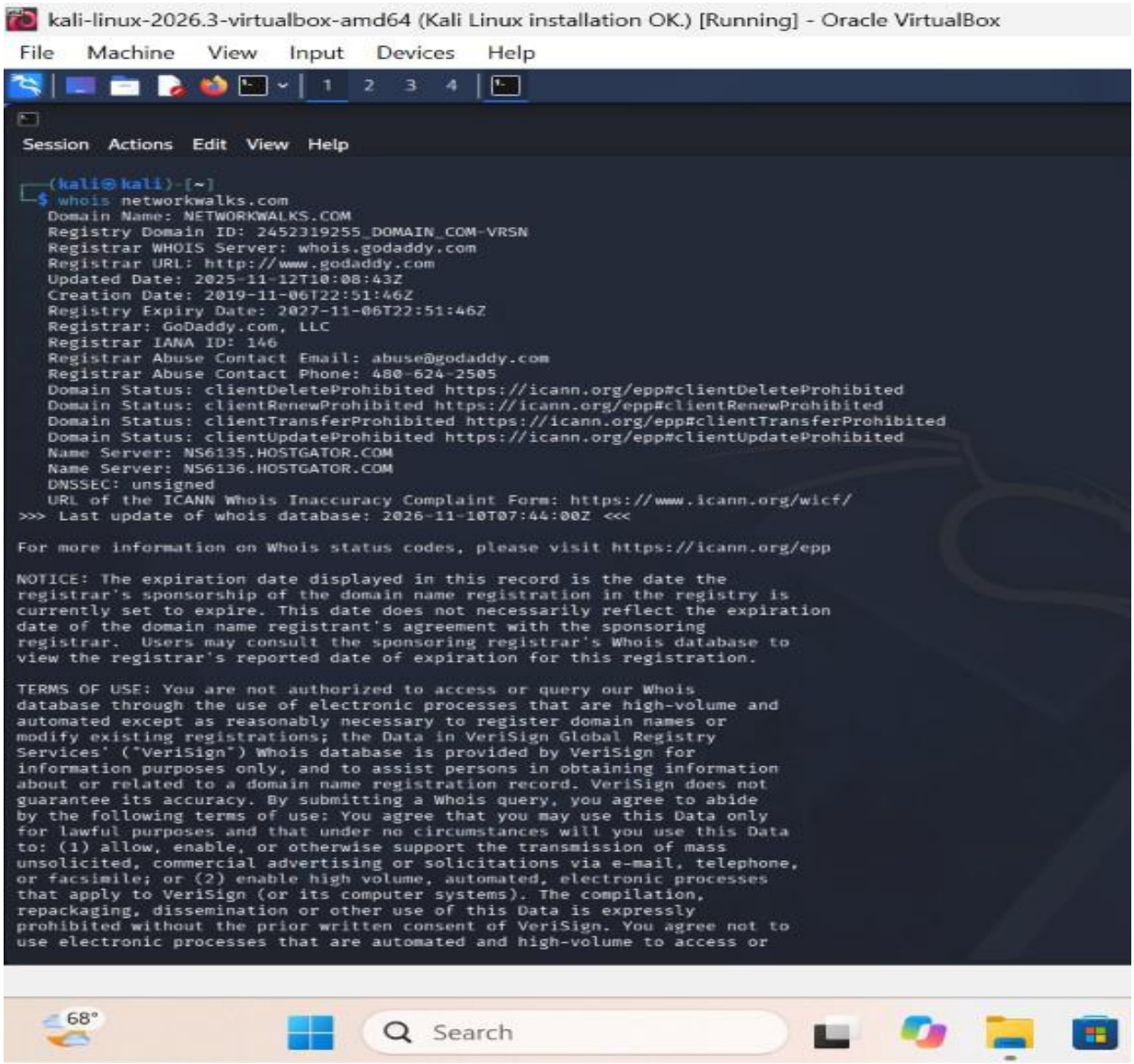


Figure 5. WHOIS command output captured in Kali Linux.

14. WhatWeb Analysis

WhatWeb was used to identify technologies and publicly visible characteristics associated with the target website.

Key observations

- HTTP redirects to HTTPS (301).
- Server architecture recorded as Apache behind an Nginx caching layer, hosted on HostGator at 192.232.216.135.
- CMS and version recorded: WordPress 7.1.4.
- Plugin recorded: WordPress Download Manager 3.3.76.
- Frontend technologies recorded: jQuery 3.7.1, Bootstrap 7.1.4, and Google Tag Manager.
- Site title: “Networkwalks Academy”; exposed contact email: info@networkwalks.com.
- Other observations include Open Graph tags, an HttpOnly cookie identified in the notes as wpdm_client, and custom security/cache headers.

- The source notes characterize public disclosure of exact CMS and plugin versions as a minor information leak.

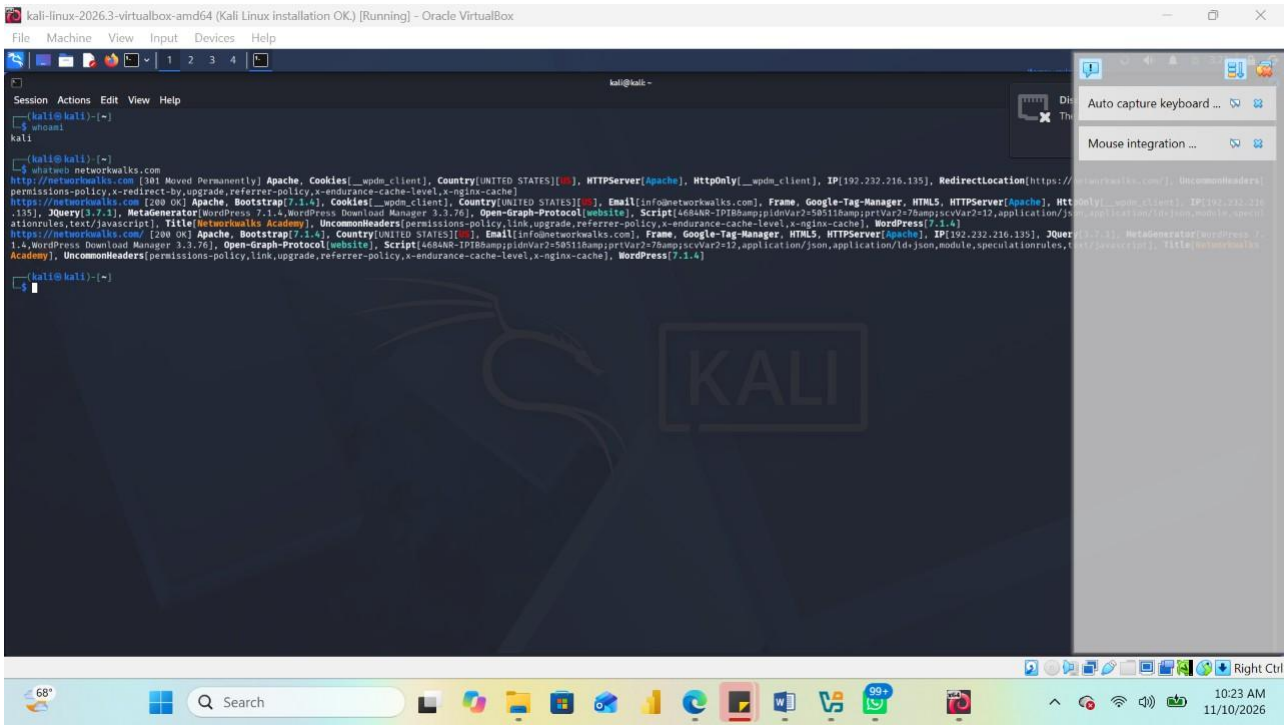


Figure 6. WhatWeb output recorded for networkwalks.com.

15. DNS Resolution with nslookup

The nslookup command was used to check whether the domain name resolves to an IP address.

Key observations

• Command recorded: nslookup networkwalks.com.

• The Kali machine used Google Public DNS (8.8.8.8).

• The domain successfully resolved to 192.232.216.135.

• The source interprets the result as successful DNS resolution and confirms the domain was live at the time of the test.

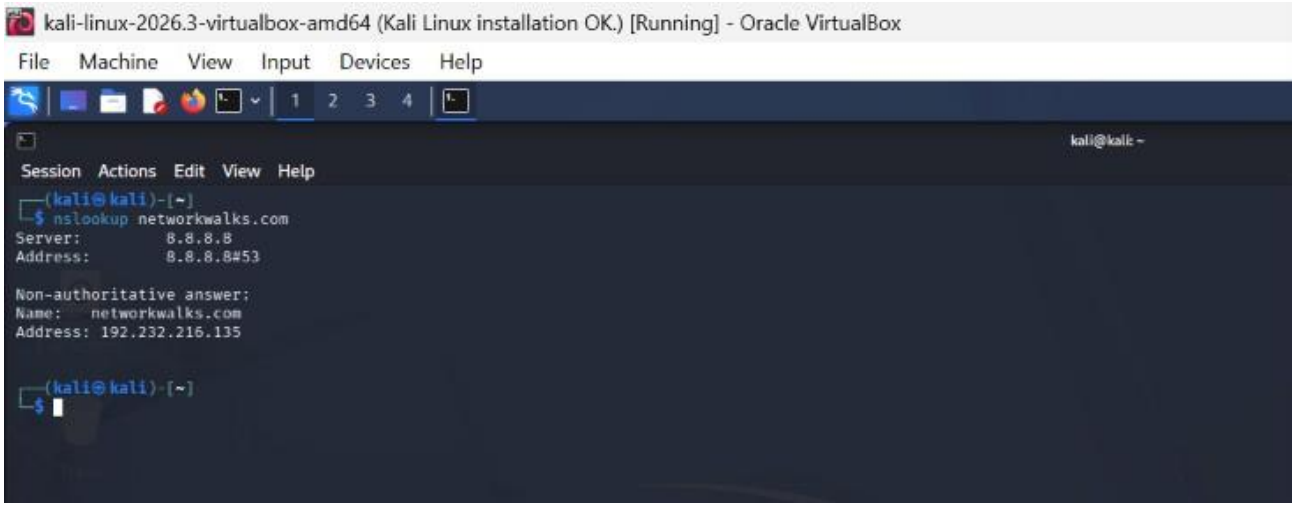


Figure 7. nslookup output showing the resolved address.

16. HTTP Header Review with curl -I

The **curl -I** command sends an HTTP HEAD request and retrieves response headers rather than the full webpage body.

Key observations

• The supplied notes record an HTTP/2 response with status 200 OK.

• The response indicates an Apache server and WordPress CMS.

• The output displayed a session cookie, security-policy headers, and the server date.

• The source materials transcribe the cookie name inconsistently; the report retains the general observation without asserting a normalized spelling.

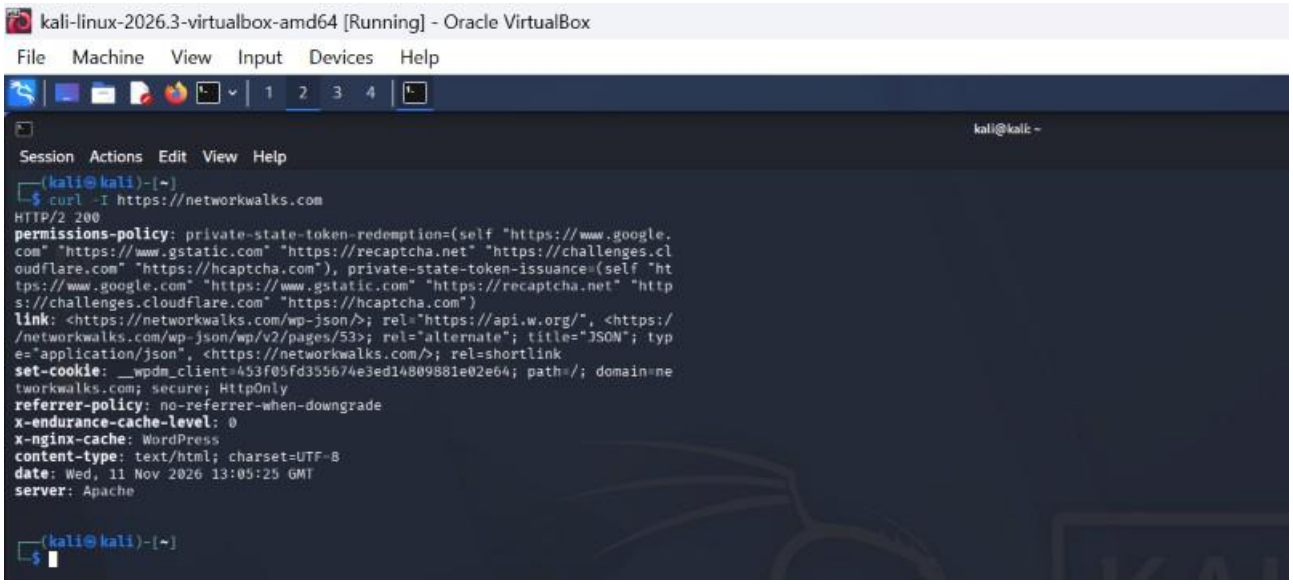


Figure 8. HTTP response headers captured with curl -I.

17. Web Application Firewall Detection with wafw00f

wafw00f was used to check whether a web application firewall could be identified in front of the target website.

Key observations

- Command recorded: wafw00f networkwalks.com.
- The tool identified ModSecurity (SpiderLabs) as the WAF.
- The screenshot and notes indicate that detection was made after two test requests.

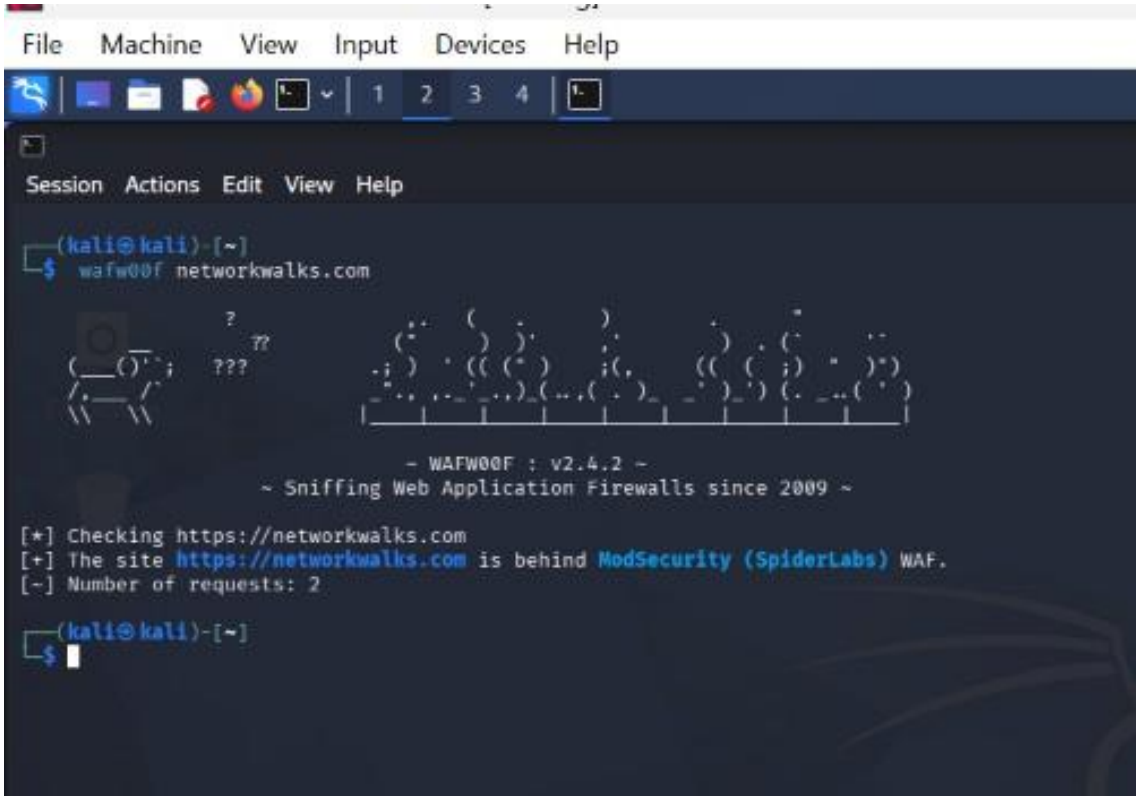


Figure 9. wafw00f output identifying ModSecurity (SpiderLabs).

18. DNS Enumeration with dnsrecon

dnsrecon was used to perform standard DNS enumeration for networkwalks.com.

Key observations

• Name servers recorded: ns61.hostgator.com and ns62.hostgator.com.

• The reference notes identify mail.networkwalks.com as the mail server (MX).

• The main website IP was recorded as 192.232.216.135.

• A zone-transfer attempt was refused by the server.

• Multiple _autodiscover SRV records were observed and interpreted in the source as indicating a cPanel hosting environment.

• The source report notes a missing SPF TXT record as a potential email-security weakness.

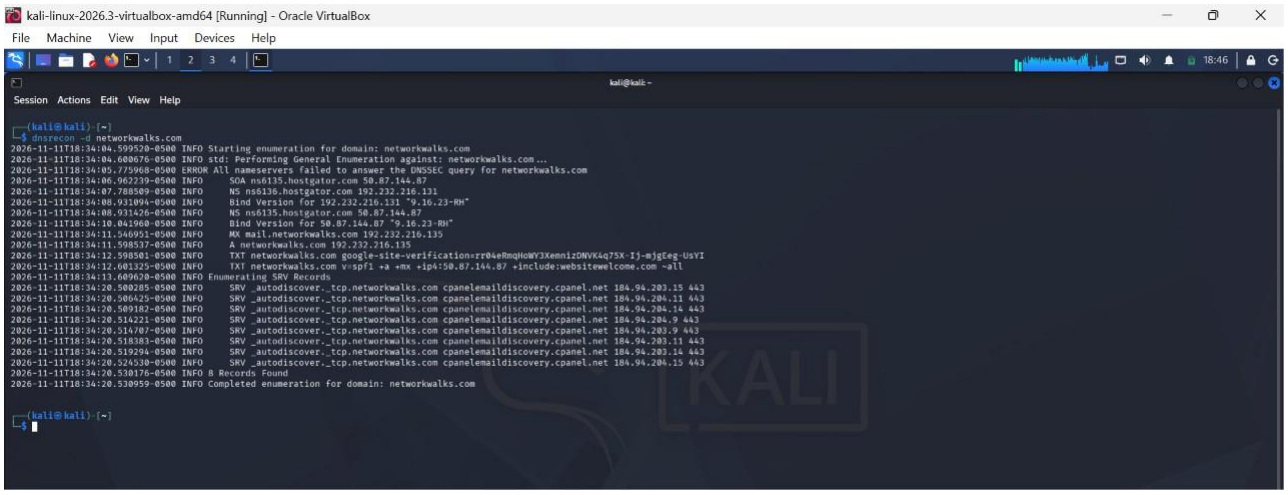


Figure 10. dnsrecon output showing DNS enumeration results.

PART B - ESSENTIALS

Week 2, Project Module: Zenmap-Based Network Scanning

Discover live hosts on the subnet using Zenmap, and view the resulting network topology.

3. Objective

The objective of this practical activity was to identify the local IPv4 configuration, determine the LAN subnet, discover live hosts on the subnet using Zenmap, and view the resulting network topology.

adapter details show an IPv4 address of 192.168.0.168, subnet mask 255.255.255.0, and default gateway 192.168.0.1. The Wi-Fi adapter description is Intel(R) Wireless-AC 9560. The document also shows a VirtualBox Host-Only Ethernet Adapter with a separate address range.

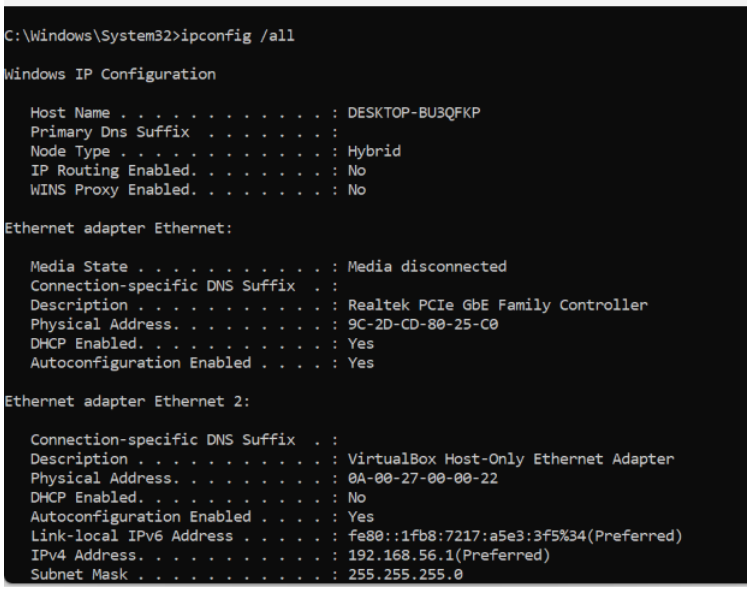


Figure 1. Windows ipconfig /all output showing adapter information and the VirtualBox host-only interface.

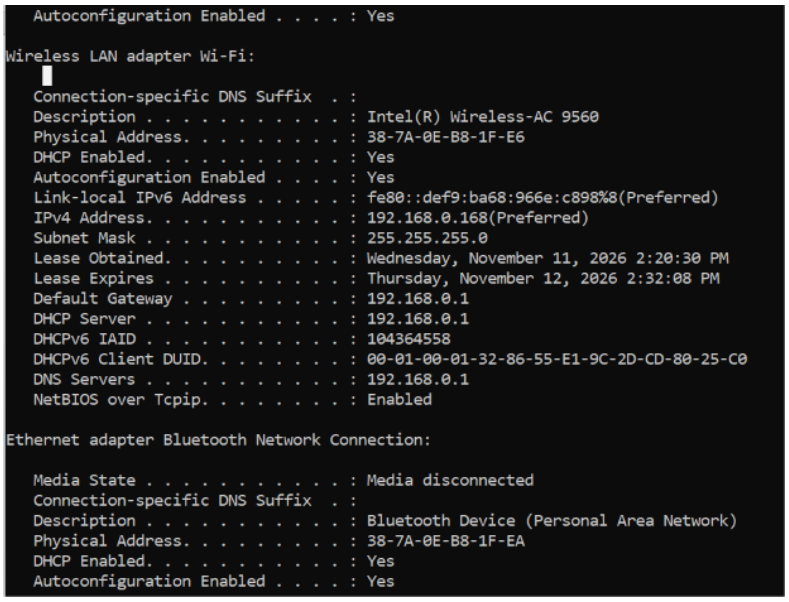


Figure 2. Wi-Fi adapter details, including IPv4 address, subnet mask, gateway, and DNS configuration.

Network addressing summary

LAN network: 192.168.0.0/24
Subnet mask: 255.255.255.0
Local Wi-Fi IPv4 shown in screenshot: 192.168.0.168
Default gateway/router: 192.168.0.1

6. Zenmap Host Discovery

Zenmap was used to perform a ping sweep of the local subnet. The command recorded in the source document was:

```
nmap -sn 192.168.0.0/24
```

The -sn option performs host discovery without carrying out a port scan. According to the supplied notes, the scan considered the 256 possible addresses in the /24 subnet and identified three active hosts.

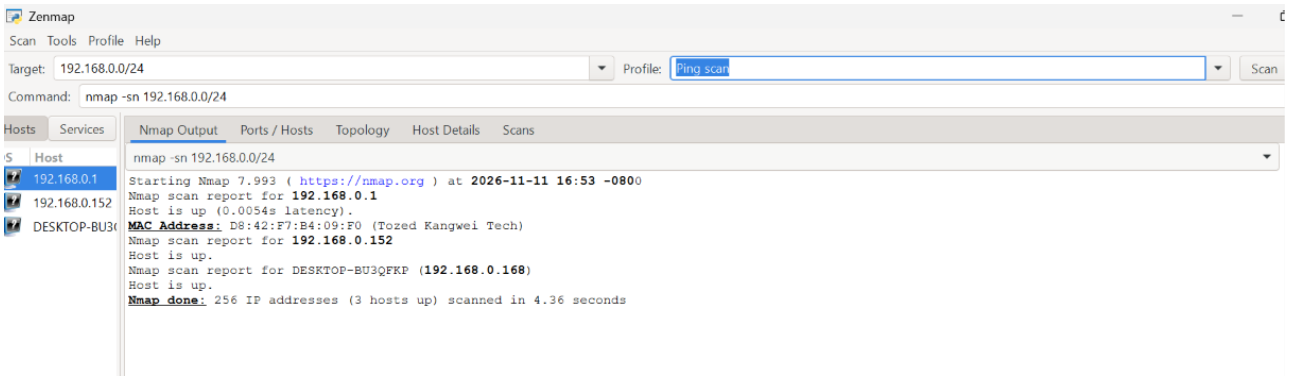


Figure 3. Zenmap output showing the host discovery scan of 192.168.0.0/24.

7. Discovered Hosts

IP address	Recorded identification	Role stated or inferred from source
192.168.0.1	—	Default router / gateway
192.168.0.152	TP-Link device	Network device
192.168.0.168	DESKTOP-BU3QFKP	Windows PC / local Wi-Fi host

The practical notes state that the three discovered hosts provide a basic map of devices connected to the physical network at the time of the scan.

8. Network Topology

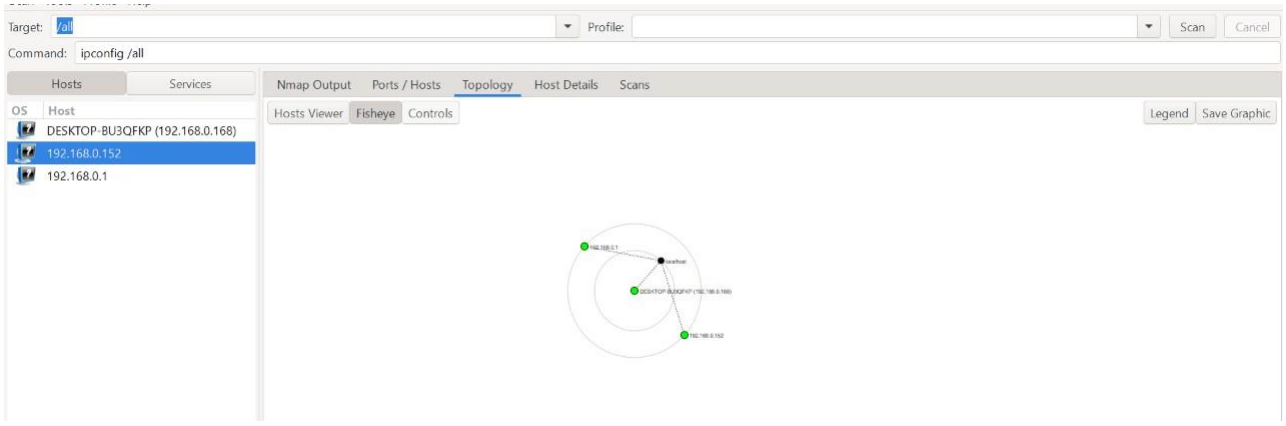


Figure 4. Zenmap topology view showing the discovered local hosts.

9. MAC Address Observations

The source document lists the following MAC addresses under the heading “MAC Address use ipconfig /all”:

• 38-7A-0E-B8-1F-E6

• B4-B5-2F-98-0C-F1

• 86-92-0D-1E-73-C3

The Wi-Fi adapter screenshot visibly shows 38-7A-0E-B8-1F-E6 for the Intel Wireless-AC 9560 adapter. The source does not explicitly map the other two listed MAC addresses to individual discovered IP addresses, so this report does not assign them to particular hosts.

10. Important Addressing Clarification

The original notes call 192.168.0.1 the “preferred IPv4 address.” However, the ipconfig screenshot identifies 192.168.0.168 as the Wi-Fi adapter IPv4 address and 192.168.0.1 as the default gateway. The Zenmap host list also identifies 192.168.0.1 as the router. This report therefore uses the screenshot-supported interpretation: **192.168.0.168 is the local Wi-Fi host address, while 192.168.0.1 is the gateway/router.**

11. Essentials Conclusion

The Zenmap-based scan successfully identified three active hosts on the 192.168.0.0/24 local network. The exercise demonstrated how to inspect local adapter settings, determine the subnet, conduct a ping sweep without port scanning, and view discovered devices in a topology diagram.

19. Electives Conclusion

The six Kali tools documented a range of publicly observable information about networkwalks.com. The WHOIS output provided domain-registration and status information; WhatWeb identified website technologies; nslookup confirmed DNS resolution; curl -I displayed HTTP response headers; wafw00f identified ModSecurity (SpiderLabs); and dnsrecon enumerated name servers and other DNS records.

20. Combined Conclusion

The two Week 2 activities demonstrate complementary cybersecurity reconnaissance skills. The Essentials activity focused on the local network: inspecting IP configuration, discovering active hosts, and visualizing the LAN using Zenmap. The Electives activity focused on external footprinting: identifying domain, DNS, web technology, HTTP, and WAF information using six Kali tools.

21. Recommendations and Good Practice

• Perform network discovery only on networks for which you have permission.

• Keep a clear record of commands, scan scope, timestamps, and screenshots for repeatability.

• For the local network, confirm device ownership and IP/MAC mappings with the network administrator or router inventory.

• For the external target, treat technology and DNS observations as reconnaissance indicators, not proof of a vulnerability or compromise.

• The source report flags public software-version disclosure and the reported absence of an SPF TXT record for review by the domain administrator.

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